

A STUDY OF THE MATRIX OF FITTING OF A SERIES OF DISCRETE
FREQUENCY FUNCTIONS ANALOGOUS TO THE TYPE A SERIES

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Introduction and Method

Frequency data not far from the normal can be fairly represented by a Type A series, which often is a normal frequency function plus corrective terms involving Hermite polynomials. Frequency data can also be fitted by a binomial frequency function $\binom{n}{x} p^x q^{n-x}$ plus corrective terms involving Krawtchouk orthogonal polynomials (see Gonin 1961).

In the particular cases where $p=q=\frac{1}{2}$, we would have the binomial function $C_x = \binom{n}{x} (\frac{1}{2})^x (\frac{1}{2})^{n-x}$ as basic frequency function plus

corrective polynomials. For the general binomial $\beta_1 = (q-p)^2 / npq$, and $\beta_1 = 0$ when $p=q=\frac{1}{2}$ and this binomial approaches the normal frequency function very rapidly. When the data for $x=0,1,2,\dots,n$, is given and n is not too large and the values of the data tend to zero at both ends, this series is especially appropriate for fitting.

A fitting by the symmetrical polynomials was done by Greenleaf (1932) and Gonin (1945), the latter using Aitken's method.

Let us graduate a set of data U_x by

$$(1) \quad y_x = [a_0 G_0(x) + a_1 G_1(x) + \dots + a_r G_r(x)] C_x$$

where $G_r(x)$ is the r th Krawtchouk polynomial given by

$$C_x G_r(x) = \frac{1}{2^n} \Delta^r \binom{n-r}{x-r} \text{ with orthogonal properties}$$

$$\sum_{x=0}^n C_x G_s(x) G_r(x) = 0, \quad s < r$$

$$= 1 / \binom{n}{r}, \quad s = r.$$

Let us now introduce the elegant orthogonal functions of Milne (1949) defined by $H_r^n(x) = \Delta^r \binom{n-r}{x-r}$ into (1). Then the data U_x can be approximately represented by

$$(2) \quad U_x \approx y_x = \frac{1}{2^n} [a_0 H_0(x) + a_1 H_1(x) + \dots + a_r H_r(x)]$$

where we drop the index n for brevity.

If we multiply (2) by $H_x(s)$ and sum over all the values of x , we obtain

$$\sum_{x=0}^n U_x H_x(s) \approx a_s \sum_{x=0}^n H_x(s) H_s(x)$$

$$\text{e.g. } C_{43} = \sum_{x=0}^3 H_3(x)H_x(4) = (1)(35)+(1)(-5)+(-3)(-5)+(-3)(3) \\ = 36$$

This produces the matrix of fitting

$$C = \begin{bmatrix} 64 & 20 & 0 & -4 & 0 & +4 & 0 & -20 \\ 140 & 64 & 20 & 0 & -4 & 0 & +4 & 0 \\ 0 & 60 & 64 & 36 & 0 & -20 & 0 & 84 \\ -140 & 0 & 60 & 64 & 36 & 0 & -20 & 0 \\ 0 & -20 & 0 & 36 & 64 & 60 & 0 & -140 \\ 84 & 0 & -20 & 0 & 36 & 64 & 60 & 0 \\ 0 & 4 & 0 & -4 & 0 & 20 & 64 & 140 \\ -20 & 0 & 4 & 0 & -4 & 0 & 20 & 64 \end{bmatrix}$$

This matrix is centrosymmetric.

The sum of the elements in the columns are each = 256. These matrices are easily computed owing to their symmetrical properties for each n once and for all and used for graduating moderately skew curves instead of Type A curves or even certain types of Pearson curves. If n becomes too large, however, the computation of the matrices become laborious.

No moments of the data have to be calculated. The data can simply be entered on the edge of a slip of paper and slid down the matrix so that inner-multiplication between the row items and data can be made and the sum ascertained and divided by 2^n to give the graduated values $y_0, y_1, \dots, y_n$.

Consider Elderton's example on page 134 of his book [see Elderton (1938)]. The graduation of a set of data has been obtained by Pearson's Type II curve, the Type A series, by Edgeworth's series and the Normal curve.

The graduation by the matrix above is simply $y = \frac{1}{128}$ CU leading to the following results;

Example

(Symmetrical curve not capable of satisfactory graduation by the Normal Curve of error)

Observations	Pearson's Type II	By matrix method	Type A	Edgeworth	Normal
11	14	15	15	16	20
116	109	103	106	106	95
274	286	287	284	285	270
451	433	445	437	436	456
432	433	432	437	436	456
267	285	274	283	284	270
116	109	107	106	106	95
16	14	19	15	16	20

In this case all the curves except the normal give excellent graduations.

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